City University of London
School of Science & Technology: Department of Engineering

EE2703 Biomedical Instrumentation
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LAB REPORT

Building an ECG
Design and Development of Biopotential Amplifier.

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Introduction

This report details the systematic design and development of an Electrocardiogram (ECG) amplifier within the scope of the Biomedical Instrumentation laboratory series, a pivotal instrument in medical diagnostics. The foundational stage of the laboratory work introduces critical components, namely the operational amplifier, differential amplifier, instrumentation amplifier, band-pass filter, and the ECG amplifier itself. Each component plays a distinct role in accurately recording and analysis of electrocardiogram signals.

Operational amplifiers serve as the primary element of the ECG amplifier, tasked with enhancing the low-voltage ECG signals to a discernible magnitude for detailed analysis. Their adaptability in configuration renders them indispensable for signal interpretation. Differential amplifiers contribute significantly to signal clarity by amplifying the variance between dual inputs and minimizing pervasive common-mode noise. Instrumentation amplifiers excel in amplifying subtle biological signals, distinguishing the crucial heart signals amidst electrical noise. The band-pass filter is integrated into the ECG amplifier to refine signal quality by attenuating extraneous frequencies.

In addition, by using an oscilloscope, we can observe the real-time effects of each component within the ECG amplifier system. This hands-on analysis allows us to compare theoretical predictions against practical performance directly. We provide an actual tangible understanding of signal gain, filtering, and noise reduction concepts. The oscilloscope's ability to visually display voltage variations over time is instrumental in verifying the functionality of operational amplifiers, differential amplifiers, instrumentation amplifiers, and band-pass filters. It also aids in fine-tuning these components to achieve the optimal signal quality necessary for accurate ECG interpretation.

Collectively, these components do not merely solidify the theoretical underpinnings but also offer invaluable practical perspectives on the intricacies of ECG amplifier design and functionality.

Laboratory equipment used:

1. Resistors, Capacitors, Op-amps
2. Breadboard
3. Signal generator
4. Oscilloscopes
5. Digital Multimeter
6. DC power supply

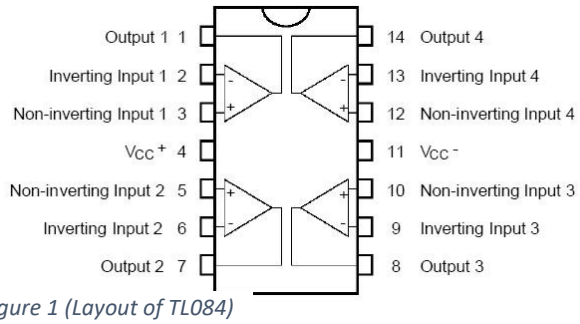


Figure 1 (Layout of TL084)

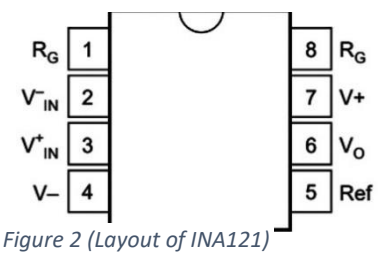


Figure 2 (Layout of INA121)

Methods

To understand how we tested and built our circuit from each block, working toward an ECG amplifier. We must understand how our power supply, oscilloscope, and breadboard were set up.

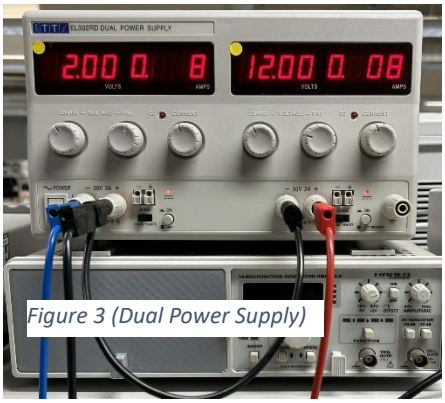


Figure 3 (Dual Power Supply)

We begin with our dual power supply in Figure 3. We need to create a ground, and we've done that by connecting two different power supply's positive and negative terminals, creating a common point serving as a reference ground for the circuit. At the same time, the unconnected terminals serve as the positive and negative voltage for the op-amps. We connect these to the breadboard using the pin and wire them into the Op-amp.

The function generator within the oscilloscope generates various waveforms; we'll be utilizing its sine waveform as its able to replicate that of an ECG signal. A digital-to-analog converter (DAC) transforms the digital representation of the desired waveform produced by the generator into an analog signal. The function generator amplifies this analog signal before sending it out as the output. The function generator can generate a broad range of signals for testing and analysis because the waveform's frequency, amplitude, and shape can typically be changed using a knob on the oscilloscope. Allowing us to manipulate the waves to test and analyze our circuits. We've also utilized software provided by the university using Keysight screen capture, allowing us to screenshot what was displayed on our oscilloscope. From now on, we'll be referring to the yellow wave as our input wave and the green wave as our output wave.

The breadboard has nodes that allow banana plugs to be inserted and holes for wires to connect it to the breadboard pins; we've combined our circuits like this. Please refer to Figure 5.

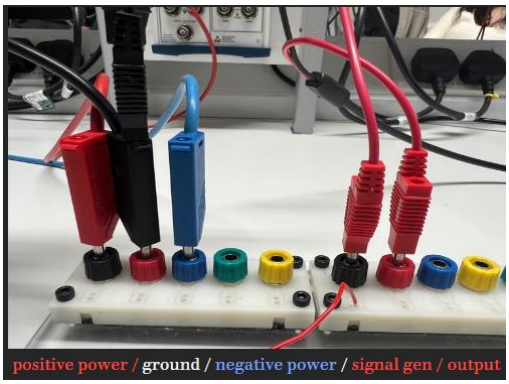


Figure 4 (Breadboard With Banana Plugs)

Lastly we use a multimeter to verify the power/voltage we're inputting to our circuits. To use a multimeter to check a power supply's voltage, first set it to measure DC voltage. After that, attach the multimeter probes to the power source, placing the black probe on the negative or ground terminal and the red probe on the positive terminal. Examine the voltage shown on the multimeter and compare it with the power supply's anticipated output value. This procedure makes sure your circuit is receiving the proper voltage from the power source, which is essential for the efficient operation of electronic parts.

A. Operational Amplifier as an Inverting amplifier

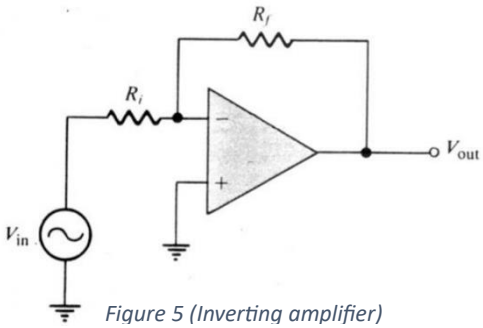


Figure 5 (Inverting amplifier)

The operational amplifier plays a crucial role in ECG circuitry, which amplifies the signals. As electric signals sent from the heart are only a few millivolts, we need to strengthen these signals so that we're able to analyze them. Furthermore, the Op-Amp is also used for isolating the patient from the electronic circuit, enhancing safety; this is necessary to prevent any risk of shock when used in a hospital environment. As this is an inverting amplifier, the output of the signal will be 180 degrees out of phase with the input signal.

To achieve a gain of 10, meaning to multiply the electric signal's input 10 times we need to select the appropriate resistor values.

The formula to calculate the gain for an inverting Op-amp is given as:

$$Gain(Av) = \frac{Rf}{Ri} = \frac{10{,}000\Omega}{1{,}000\Omega} = 10$$

Therefore, in the experiment, the values we selected for Rf and Ri were.

Rf = 10,000Ω and Ri= 1,000Ω

We Connect the +15V and –15V into the tl084 for pins 4 and 11 (shown in figure 1). Connect it with a wire under the banana pin of the wave generator and out into pin 2 to get an inverting amplifier.

We selected 1V amplitude Pk-Pk; this is large enough to ensure the signal is clear enough for observation. At the same time, we are avoiding the risk of pushing the Op-amp to saturation.

Q7) However, the gain we obtained from our trial was not 10. Instead, we achieved a gain of Av = 10.1/1.05 = 9.62 (dB) using values from (Figure 7).

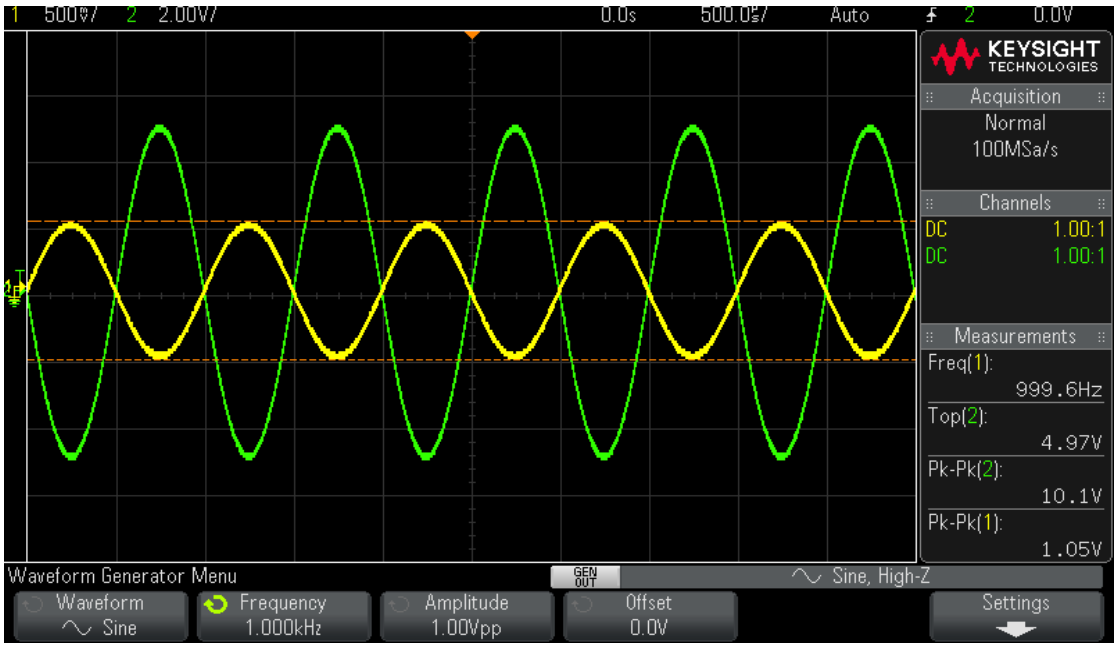


Figure 6 (I.A. Wave at 1kHz and 1.00Vpp Amplitude)

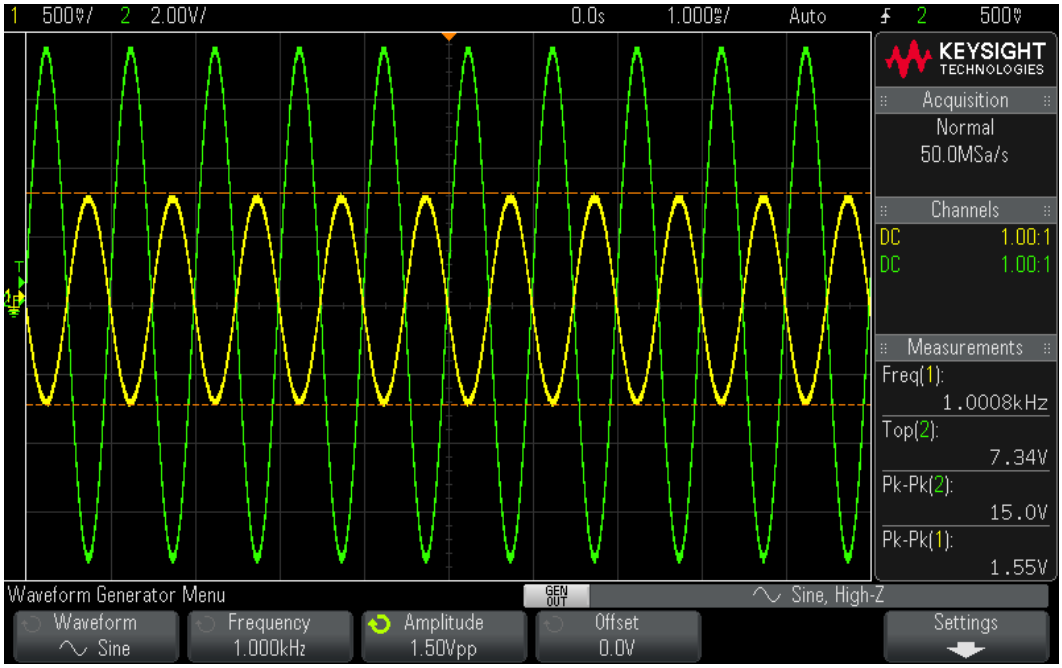


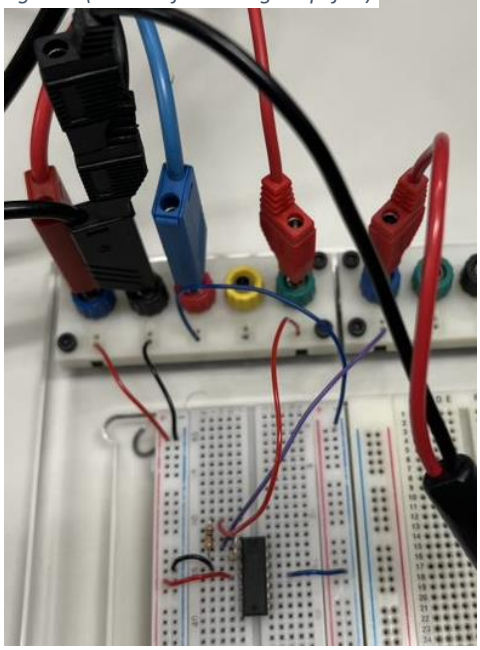
Figure 7 (I.A. at 1KHz and 1.5p-p Amplitude)

Q8) Gradually increasing the input voltage would increase the output proportionally. However, in the opposite direction, it's an inverting amplifier. In (Figure 8) our gain slightly increased as Gain = 15.0 V / 1.55 V = 9.68 dB (2 d.p.) by 0.06. However, anything over 3V amplitude Pk-Pk would lead the op-amp to reach saturation. The output signal experiences clipping but cannot increase any further, meaning the portion of the signals that would exceed the limit is cut off, leading to distortion of the waveform. This helps verify that we are only inputting 15V into our op-amp.

Q9) Reselecting our resistor values would produce a different gain; we have a gain of 0.5. Therefore, we need the value of Ri to be larger than Rf. We've selected values of:
Rf = 10,000Ω and Ri = 5,000Ω.

$$Gain = \frac{Rf}{Ri} = \frac{10{,}000\Omega}{5{,}000\Omega} = 0.5$$

Figure 8 (Circuit of Inverting Amplifier)



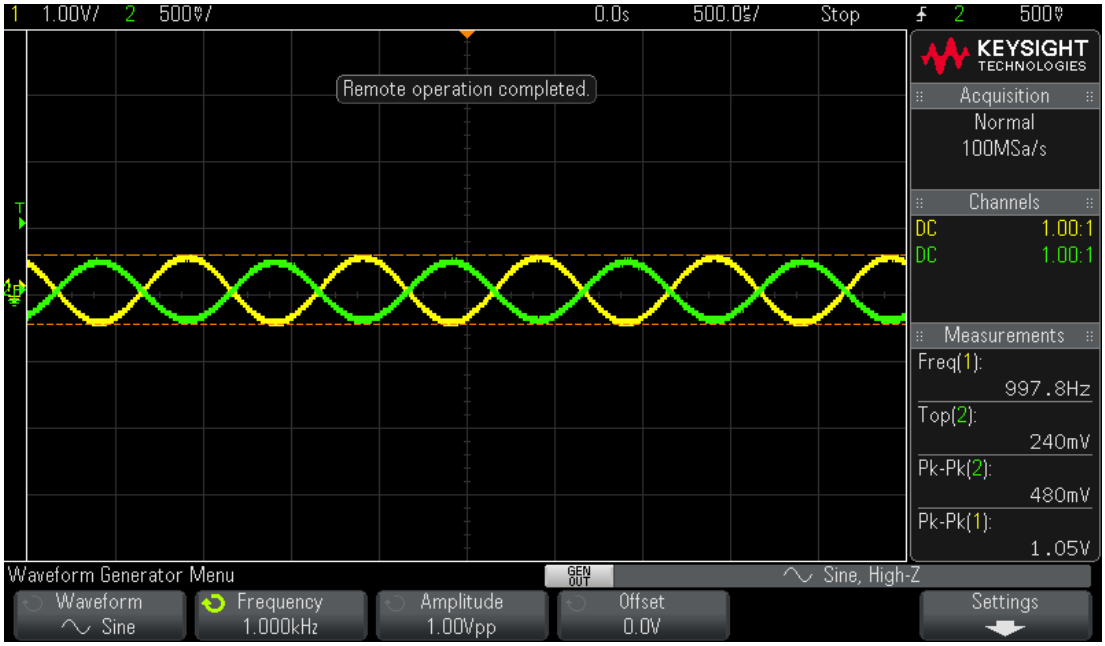


Figure 9 (I.A. With 0.5 Gain at 1KHz)

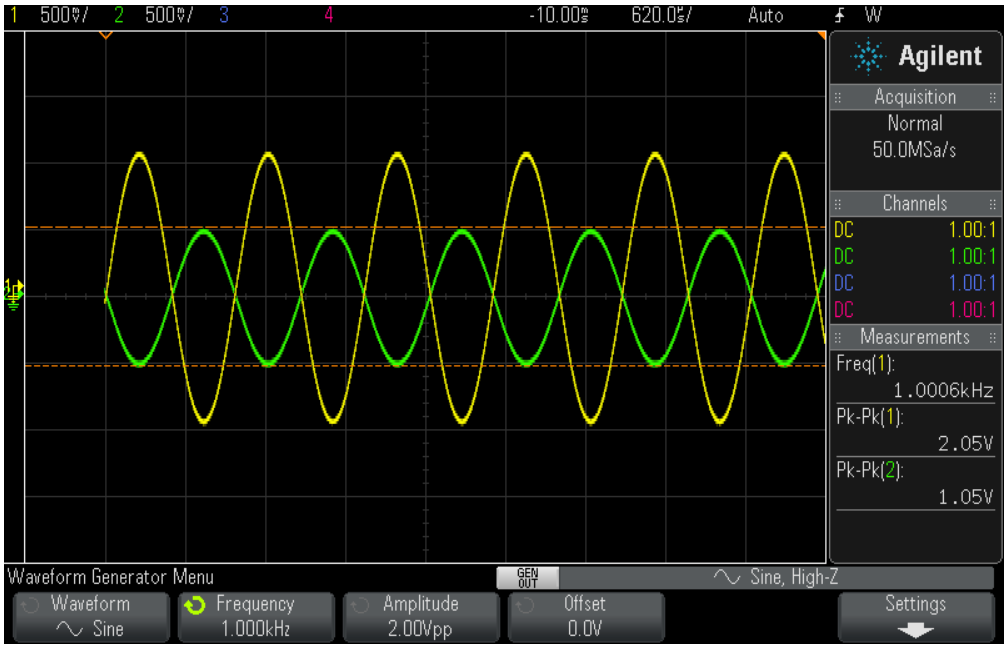


Figure 10 (I.A. With 0.5 Gain at 1KHz and 2Vp-p)

As you can see above in (Figure 10) our gain = 480 mV / 1.05 V = 0.46. Increasing its Pk-Pk amplitude to 2.0V showed a change in the gain. Gain = 2.05V / 1.05V = 0.51 dB Using values from (Figure 11). Our gain has increased slightly and exceeds that of our theoretical value by 0.01 dB, corresponding to our previous experiment with a gain of 10. If the inputs of our inverting amplifier were switched, it would no longer be an inverting amplifier. It would become a non-inverting amplifier, meaning the output sine wave signals observed previously would be in phase with the input sine waves. Since our inverting amplifier's output sine waves were 180 degrees out of phase in respect to the input signal.

B. Differential amplifier

A crucial component in ECG amplification, it helps eliminate noise by differentiating what is common at both inputs and eliminating it from the signal while amplifying the difference between the two inputs/electrodes. This helps isolate the signal, essential for detecting the tiny electrical changes caused by the heart. Therefore, creating a clear and concise ECG reading generated by the heart's biopotential reduces the signal-noise ratio of the reading. While also allowing for flexibility in the various pairs of electrode placement.

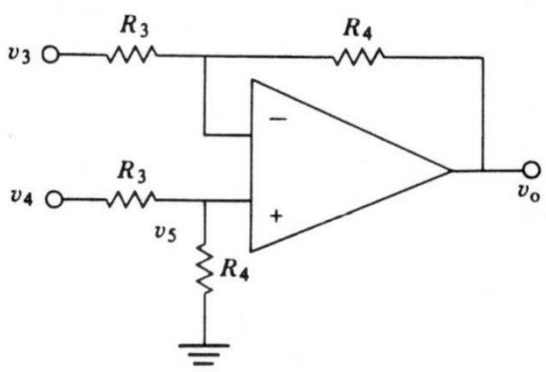


Figure 11 (Differential Amplifier)

The differential gain is the amplification factor applied to the difference in voltage between the two input signals. It allows the amplifier to amplify the weak ECG signal we're interested in while minimizing the noise. The formula is given below:

$$Theoretical\ Differential\ Gain\ (Gd) = \frac{R4}{R3}$$

Common-mode gain is the amplification factor given to signals that are common to both signals such as electrical interference from equipment, and signals arising due to movement and breathing.

$$Common\ mode\ gain\ (Gc) = \frac{Vout}{Vin}$$

CMRR (Common-Mode Rejection Ratio) measures how well an amplifier can reject common signals between the two inputs relative to differential signals. A high CMRR means that the amplifier is good at amplifying the signals of interest (differential signal) whilst ignoring the interference (common-mode signal). A high CMRR is essential for equipment such as the ECG for accurate assessment and diagnosis of the patient.

$$CMRR(dB) = 20log\frac{Gd}{Gc}$$

Q1) To obtain a differential gain of 18 we selected values R4=18,000 Ω and R3= 1,000 Ω. Both the resistors are connected to the Inverting and non-inverting input.

$$Gd = \frac{18{,}000}{1{,}000} = 18$$

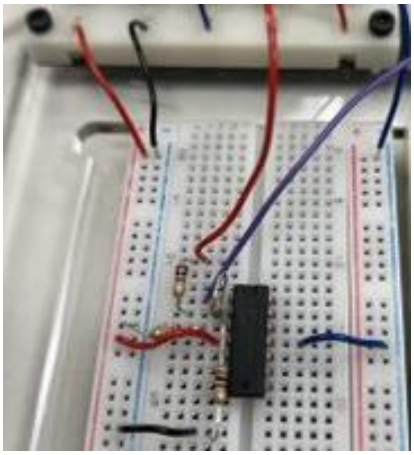


Figure 12(Built Differential Amplifier)

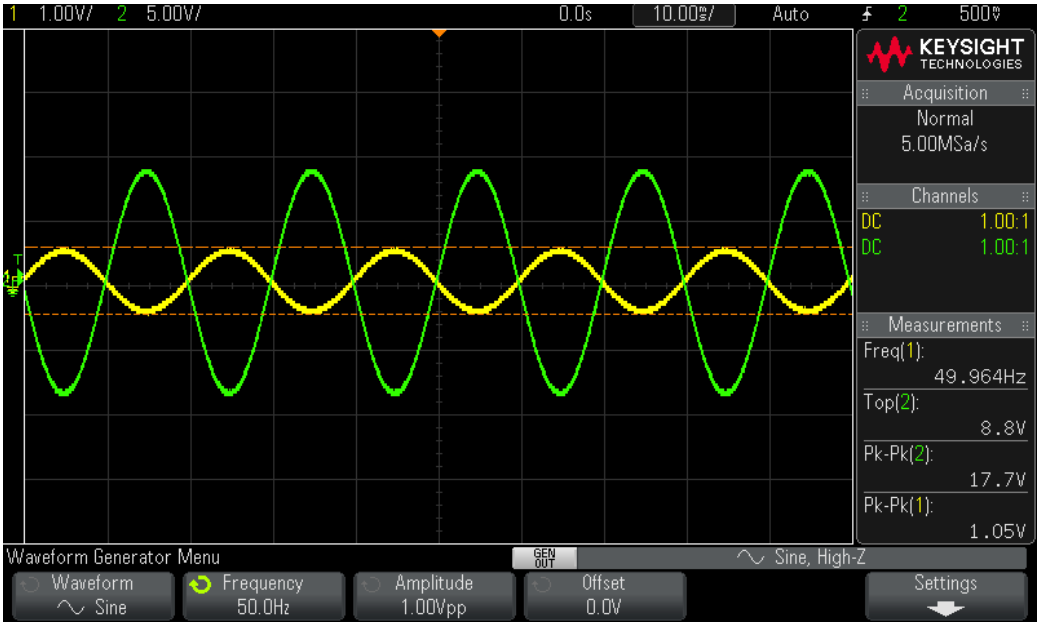


Figure 13 (D.A. at 50Hz and measuring Differential Gain)

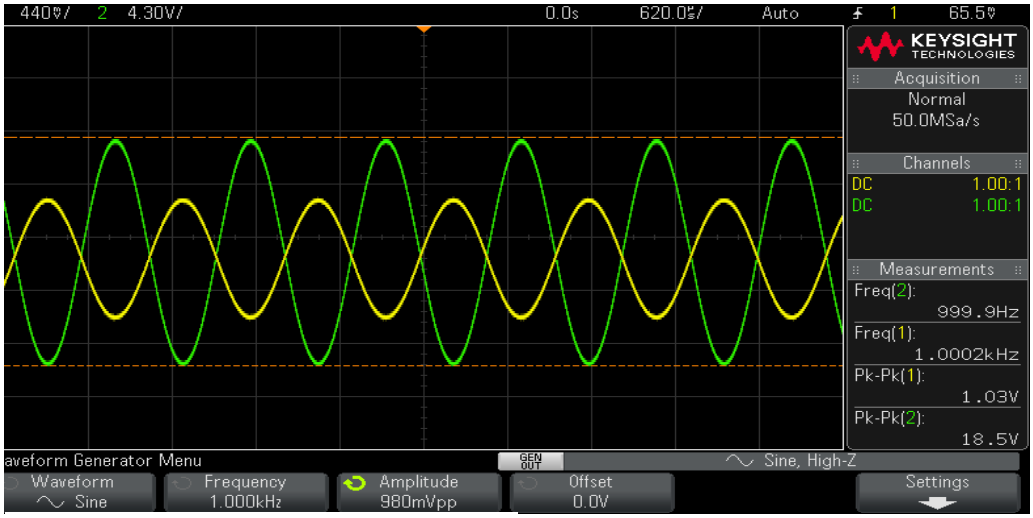


Figure 14 (D.A. at 1KHz measuring Differential Gain)

Q6) As one of the inputs is grounded, it helps ensure the input signal is purely differential and eliminates the common-mode signal component. The output observed on an AC-coupled oscilloscope then directly corresponds to the amplification of the input signal, providing a clear and straightforward means to calculate the differential gain. At 50Hz (figure 14), our differential gain is Gd = 17.7 V / 1.05 V = 16.86, which is 1.34 less than our theoretical gain. Taking values from a much higher frequency (figure 15), Gd = 18.5 V / 1.03 V = 17.96, which is 0.04 less than our theoretical gain.

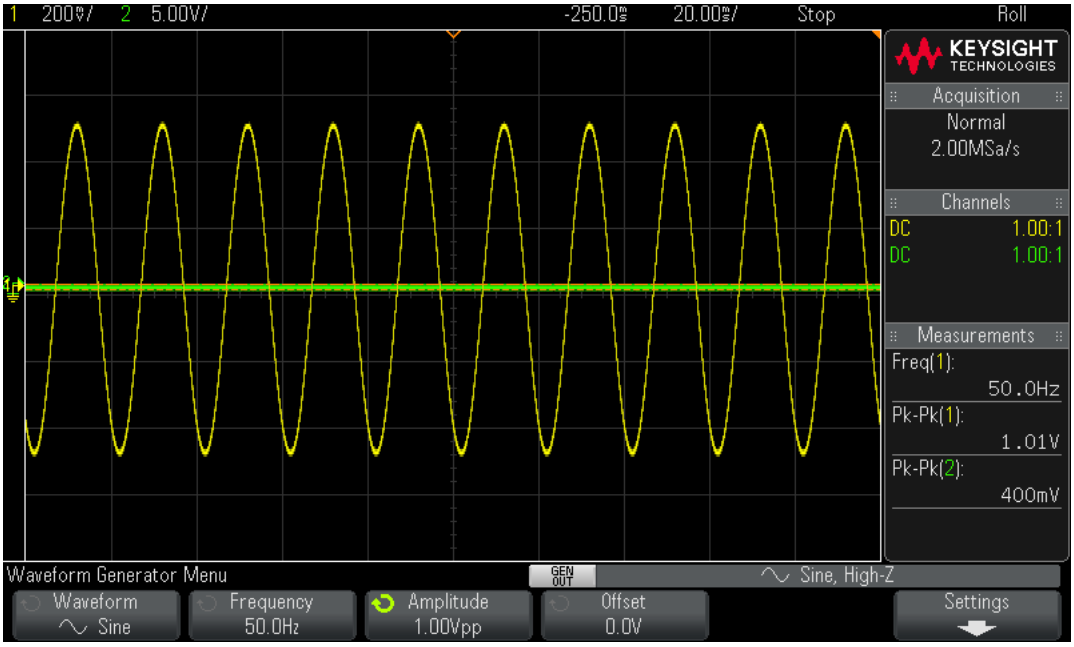


Figure 15 (D.A. Measuring Common Mode Gain)

Q7) To measure the common-mode gain, we drive both inputs together by connecting both V1 & V2. This setup ensures that the same signal is applied to both inputs. By doing this, we can measure how much the identical signal is amplified. Giving the common mode gain, which is 0.396, indicates fair noise rejection.

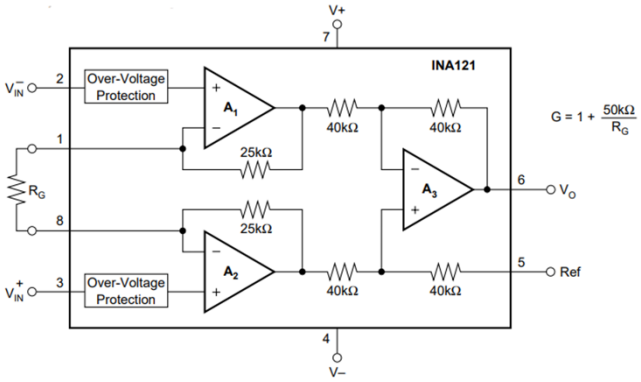
$$Gc = \frac{400mV}{1.01\,V} = 0.396$$

Q8) We are analyzing our 50Hz values. Ideally, we'd want the common mode gain to be 0, and theoretically, it should be 0. We obtained the value 0.396 dB, 0.396 higher than our theoretical values. For our differential gain value, we have obtained a value of 16.86 dB which is 1.14 lower than our theoretical gain.

Q9) When calculating our CMRR (Common Mode Rejection Ratio), we've obtained differential gain and common mode gain at 50hz, allowing us to calculate it, which is 32.58, which is a fair CMRR value but not high. A higher CMRR value is usually more desirable, especially for medical devices like the ECG, as it ensures better noise rejection and signal fidelity.

$$CMRR = 20\log\left(\frac{16.86}{0.396}\right) = 32.58\,dB$$

C. **Instrumentation amplifier**



Instrumentational amplifiers are used in their ability to accurately amplify tiny biopotential signals from the heart while rejecting noise and interference. They have high input impedance, meaning the amplifier draws tiny current from the source signal it's measuring; this helps prevent the amplifier from affecting the signal itself, preventing signal distortion from the body's resistance, and offers excellent common-mode rejection, essential for filtering out noise that's common to both input leads. This ensures clear, precise ECG readings, which is critical for effective diagnosis and monitoring in medical applications.

For this experiment, we are not using the TL084 chip; instead, we are using the INA121 chip. This allows for adjustable gain as the gain of the amplifier can just be changed with a single external resistor. This feature is useful when the amplification is based on different signal amplitudes or focuses on specific aspects of the ECG signal.

1. We have selected a 510 Ω resistor as Rg to produce a gain of 99. 1 + 50,000/510 = 99

$$G = 1 + \frac{50k\Omega}{Rg}$$

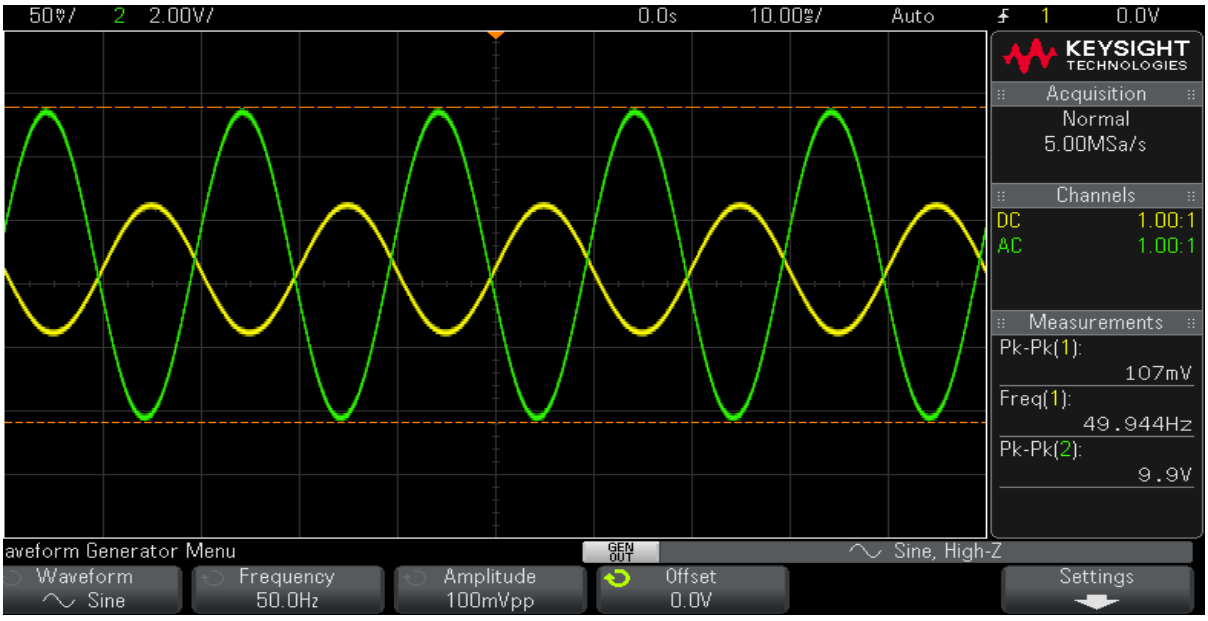


Figure 16 (Instrumental Amplification measuring differential gain)

Q5) We proceeded to measure the differential gain at 50hz (Figure 18), by connecting one of the inputs to the ground. This ensures that it's purely the differential gain we're analyzing and removing the common mode noise. We compute that our differential gain is 92.52.

$$Differential\ gain = \frac{Vout}{Vin} = \frac{9{,}900mv}{107mv} = 92.52$$

Q6) Our

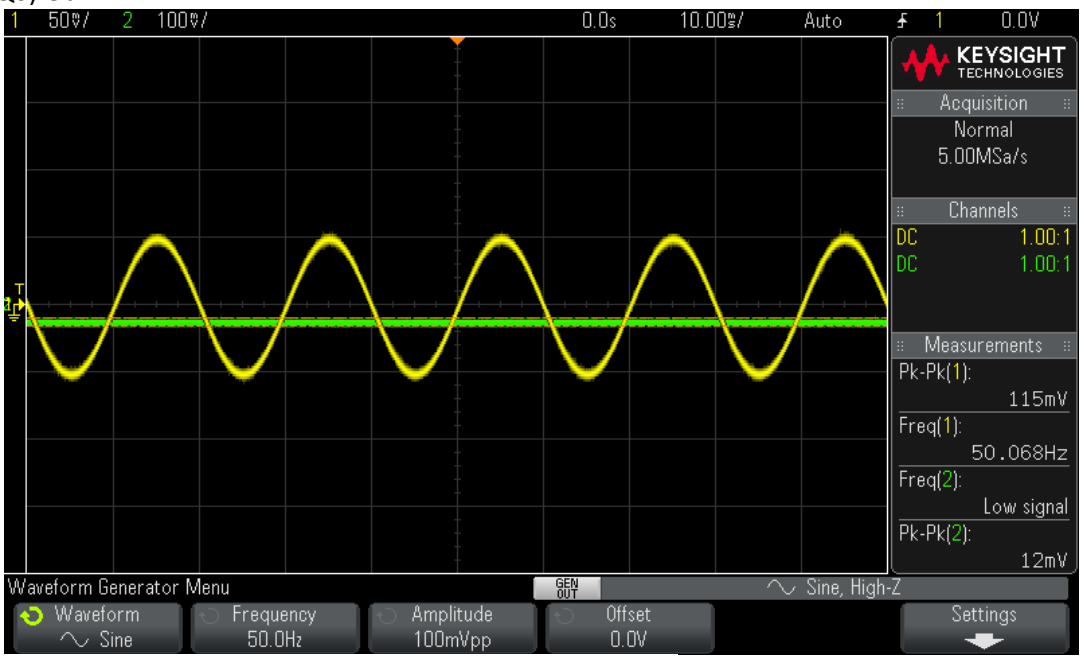


Figure 17 (Instrumental Amplification measuring common mode gain) Common mode gain is approximately 0.104 using the values in our (Figure 19) above. This is exceptionally good as a lower common mode gain means better noise rejection and allows for more precise amplification of signals of interest.

$$Common\ mode\ gain\ (Acm) = \frac{Vout}{Vin} = \frac{12mv}{115mv} = 0.104$$

Q7) To calculate the CMRR (Common Mode Rejection Ratio), we require differential and common-mode gain, which we estimated at 50Hz. Therefore, using the equation in part b, we determine that our CMRR = 58.98 dB (2 d.p.). This indicates reliable performance in the amplifier's ability to filter/reject common-mode noise and focus on only amplifying the differential signal. The higher the CMRR, the better the amplifier can distinguish essential signals from noise.

$$CMRR = \frac{Gd}{Gc} = 20\log\left(\frac{92.52}{0.104}\right) = 58.98\ dB$$

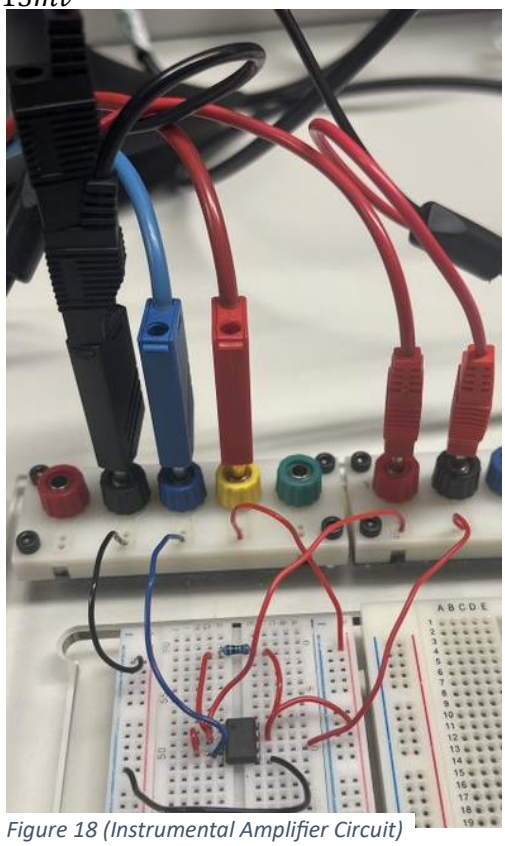


Figure 18 (Instrumental Amplifier Circuit)

D. Band-pass filter

The bandpass filter serves a dual function, simultaneously acting as a high-pass and low-pass filter. This means that only a specific frequency range would be processed into a signal; high-frequency noise, e.g. (electrical interference from devices or EMG) and low-frequency noise, e.g. (breathing or movement) would be simultaneously negated by the band pass filter. Focusing on the frequency range at which the heart's electrical signals predominantly lie allows for a more accurate and precise reading of the heart's activities. Crucial for diagnostics and monitoring, enhancing the quality of ECG reading.

We are using Butterworth's filters without any ripples in the filter's frequency response within the frequency range we are interested in. It provides a smooth and uniform transmission while effectively attenuating signals above the frequency cut-off. Butterworth filters are known for their simplicity and efficiency in applications with a desired smooth response. However, it has a poor roll-off rate (figure 22). We attempt to recreate its circuit layout and achieve the gain it has when its second order are shown below (figure 21).

The cut-off frequency of a filter is the frequency at which the filter starts to reduce the signal's strength significantly. At the cut-off frequency, the output signal is 70.7% of the input signal, where the sharp drop-off occurs. It is crucial in determining the filter's effective range: for a low-pass filter, it's where higher frequencies are attenuated, and for a high-pass filter, it's where lower frequencies are reduced. Its formula is given as: $fc = \frac{1}{2\pi R1C1}$

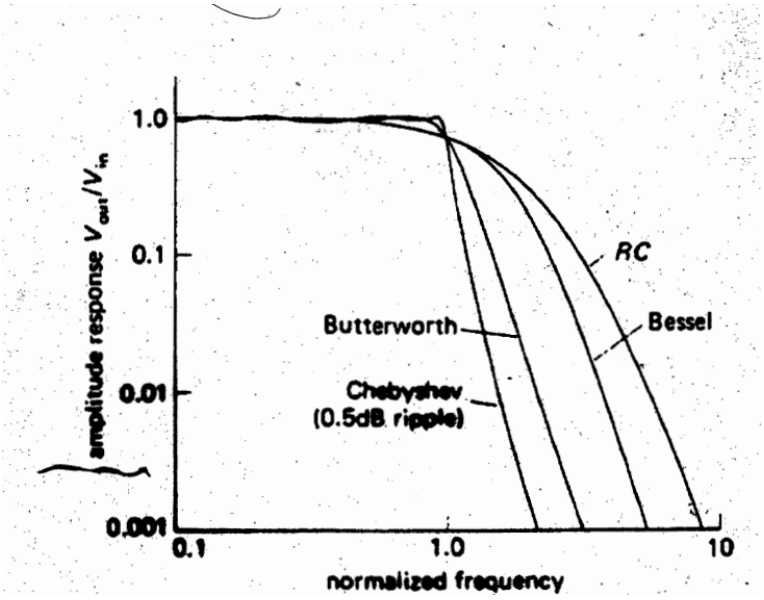


Figure 20 (Frequency to Amplitude Response)

TABLE 5.2. VCVS LOW-PASS FILTERS

Poles	Butterworth K	Bessel f_n	Bessel K	Chebyshev (0.5dB) f_n	Chebyshev (0.5dB) K	Chebyshev (2.0dB) f_n	Chebyshev (2.0dB) K
2	1.586	1.272	1.268	1.231	1.842	0.907	2.114
4	1.152	1.432	1.084	0.597	1.582	0.471	1.924
	2.235	1.606	1.759	1.031	2.660	0.964	2.782
6	1.068	1.607	1.040	0.386	1.537	0.316	1.891
	1.586	1.692	1.364	0.768	2.448	0.730	2.648
	2.483	1.906	2.023	1.011	2.846	0.963	2.904
8	1.038	1.781	1.024	0.297	1.522	0.238	1.879
	1.337	1.835	1.213	0.599	2.379	0.572	2.605
	1.889	1.956	1.593	0.861	2.711	0.842	2.821
	2.610	2.192	2.184	1.006	2.913	0.990	2.946

Figure 19 (Gain reference for different active filters)

Two Pole low pass filter (with cut-off frequency 20Hz)

Q1) Based on the capacitors we have in our Finkelstein Laboratory we selected capacitor values of C1/2 = 100nF to achieve a cut-off frequency of 20Hz we've selected R1/2 = 80k Ω. Ra = 100k Ω and Rb = 56k Ω. To achieve the gain of 1.57

$$Gain = 1 + \left(\frac{Ra}{Rb}\right)$$

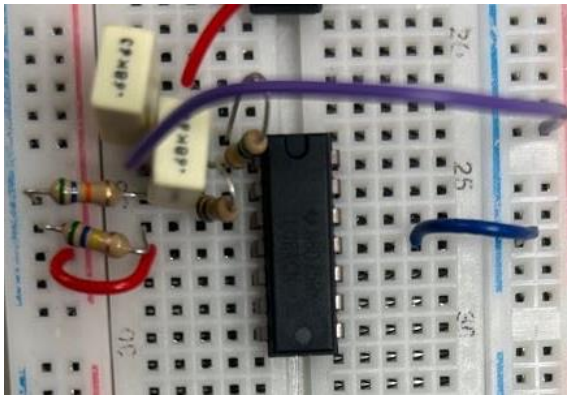


Figure 21 (2nd Order Active Low Pass

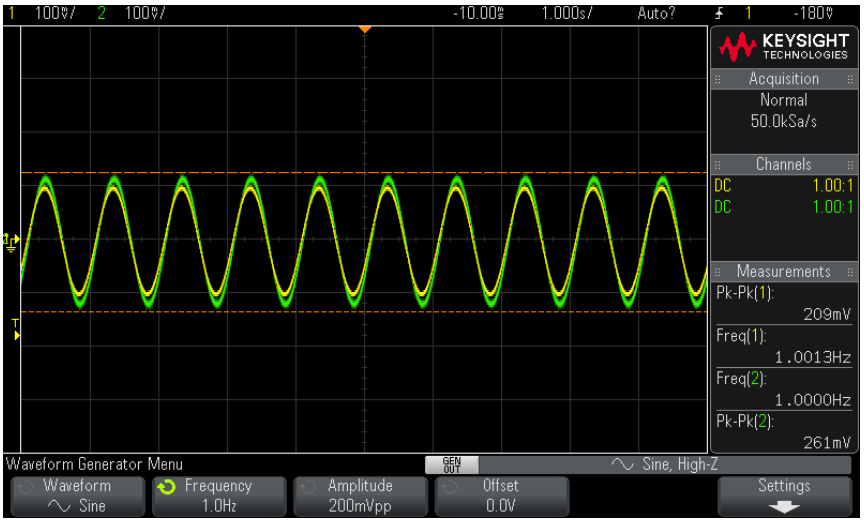


Figure 22 (Low Pass filter wave at 1Hz)

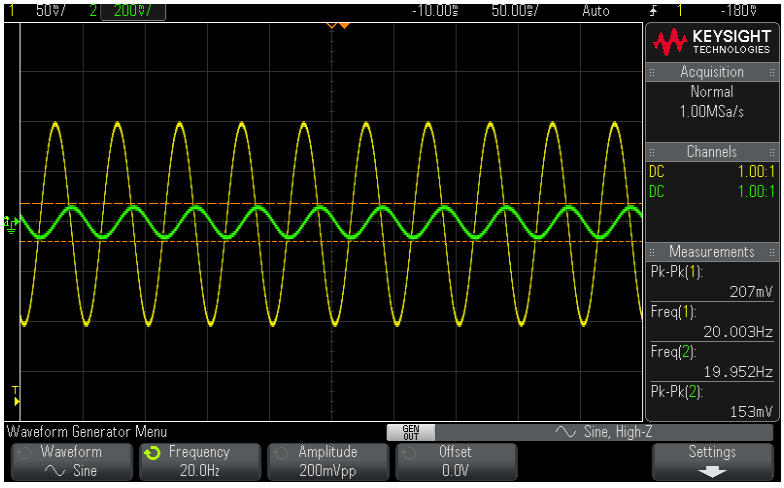


Figure 24 (Low Pass filter wave at 20Hz)

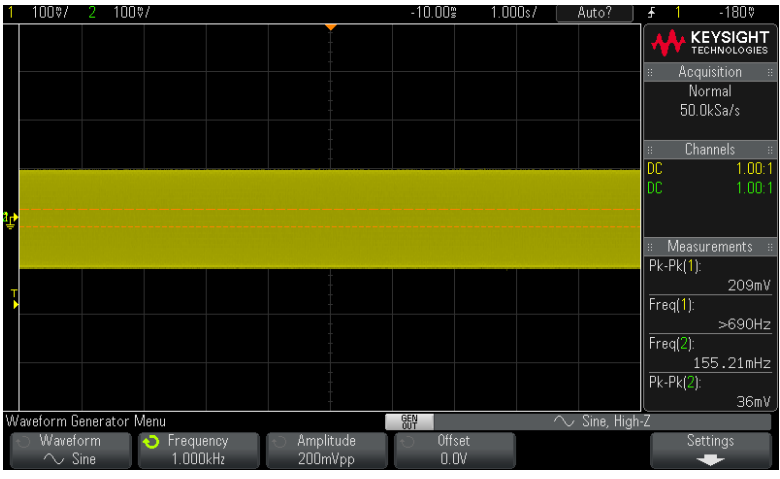


Figure 23 (Low Pass filter wave at 1KHz)

Q3) To experimentally confirm the function of the low pass filter, it would attenuate any signal above the cut-off frequency, which is 20Hz. We select a low frequency, such as 1Hz, and slowly increase it to the cut-off frequency. The figure above shows that our signal is still amplified by approximately. Gain = 261mV/209mV = 1.25.
As we reach the cut-off frequency, the slew-off rate is shown as the Gain = 153/207 = 0.73, close to the 0.707 gain we were looking for. Lastly, by increasing the frequency to 1kHz, we observe in the figure that our output sine wave has wholly disappeared, and we have gain = 36/209 = 0.17. We are proving that our low-pass filter is functional and can attenuate signals at a frequency higher than 20Hz.

Two-Pole High Pass Filter (with cut-off frequency 0.4Hz)

Q1) To achieve the gain of 0.4Hz, we have decided to use the capacitors 680nF for C1/2 and resistor values of 581.5KΩ for R1/2. These values were strategically chosen based on what capacitors and resistors we had available. Ra & Rb remain the same as it's used to achieve the gain, we want of 1.57.

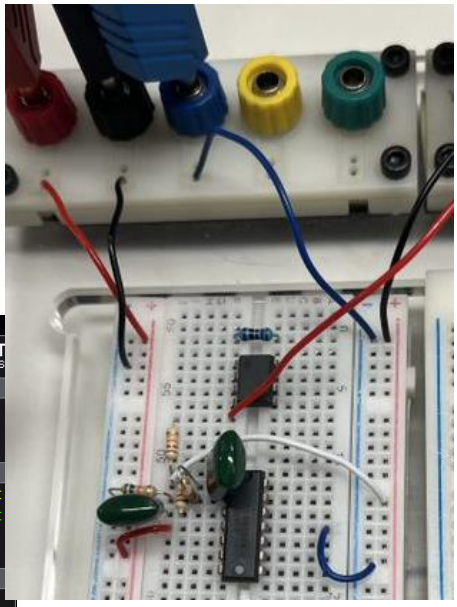


Figure 25 (2nd Order Active High Pass

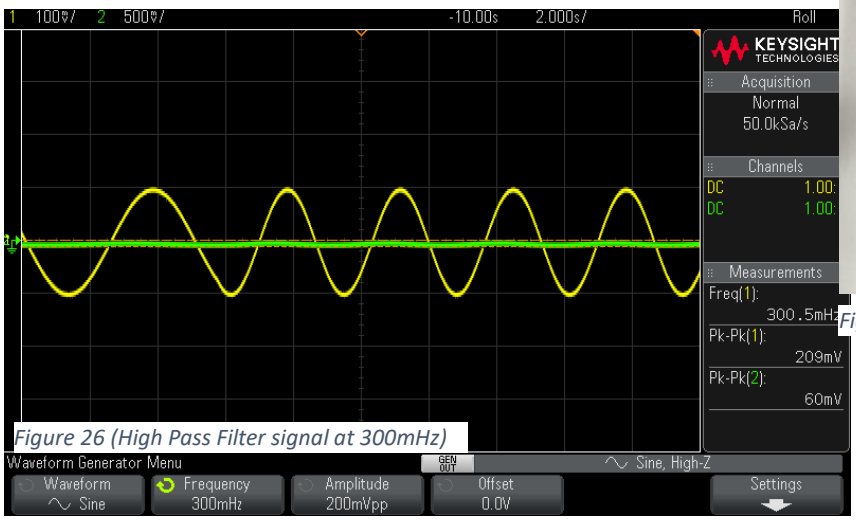


Figure 26 (High Pass Filter signal at 300mHz)

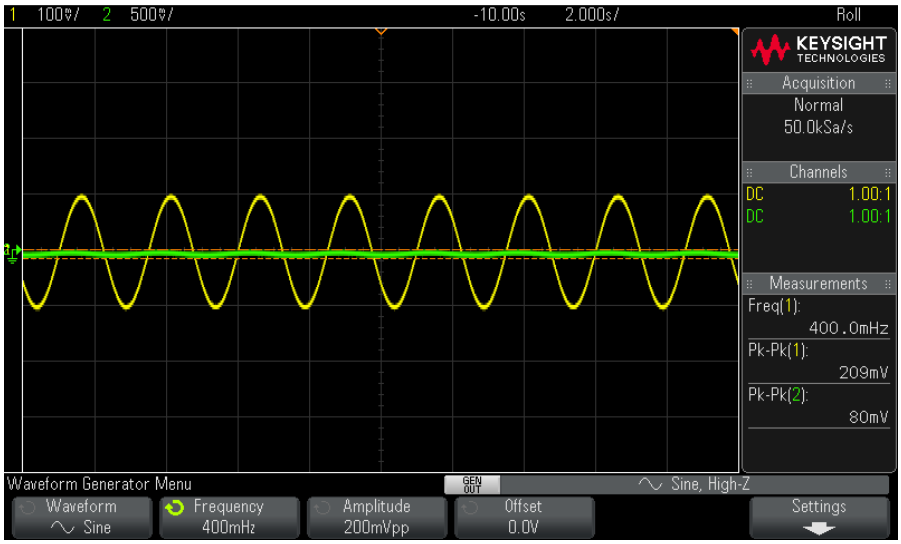


Figure 27 (High Pass Filter signal at 400 mHz)

Q3) To confirm the operation of our high-pass filter, we understand that it attenuates signals lower than 0.4Hz. So, we begin at 0.3Hz to prove that signals below 0.4Hz would be attenuated, which shows the output signal is flat and has not been amplified but filtered. It has gain = 60mV/209mV = 0.29, showing a sharp drop off. As we set our frequency to 0.4Hz (cut-off frequency), we observe that the signal has undergone a similar effect to our 0.3Hz counterpart. It's also relatively flat and has not been amplified; it has a gain = 80/209 = 0.38, not what we expected. However, as we increased the frequency, our output signal returned with a gain = 181mV/208mV = 0.87, which is still much lower than we expected. Nonetheless, it proves the function of our high-pass filter. Although not very well, it can filter out signals below the frequency of 0.4Hz.

Bandpass filter

Q1) The order in which the high-pass or low-pass filter is connected doesn't affect the bandpass. If their frequency cut-off range is not overlapping, for our bandpass filter, we have fc = 0.4 Hz and fc = 20hz.

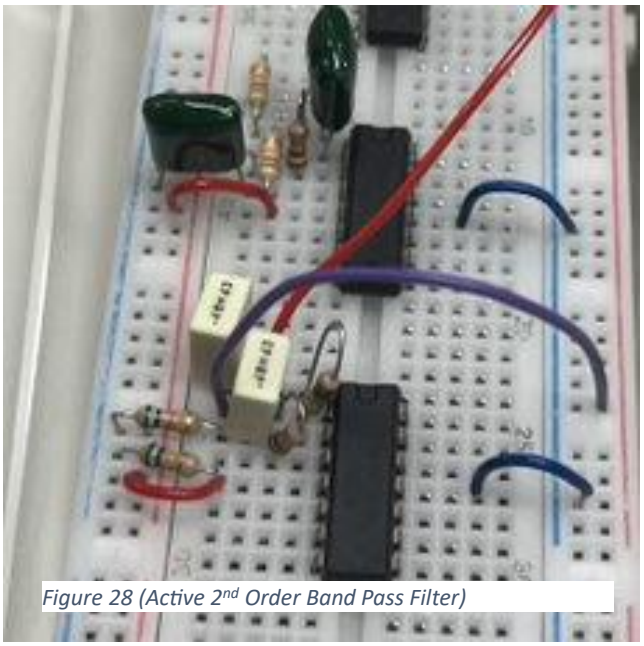


Figure 28 (Active 2nd Order Band Pass Filter)

Q4) As seen from the data below (figure 33), we have assessed the bandpass filter against a frequency range of 0.1Hz-100Hz. We expect -3dB gain for frequency cut-off from our maximum gain, which is 8.94 at the range of 5-10Hz; if we interpolate from our graph, we can figure out where our device's cut-off frequency was around. From our results table, between 0.5-0.7 Hz and between 20-22Hz is where our -3dB gain of 5.93 would be. Using linear interpolation, we figured out that our cut-off frequency for our bandpass is fc = 0.62Hz and fc = 21Hz. We will give a tolerance of ±5% for our selected components. For our low-pass filter, the range is only +1Hz off our theoretical value, which is fair. This means the component we picked for this filter is within the tolerance range of our used components. Meanwhile, the high pass filter was +0.22Hz off our theoretical value, meaning it does exceed the tolerance range of the components used. Meaning that there may have been faulty components, or we have not optimized our circuit.

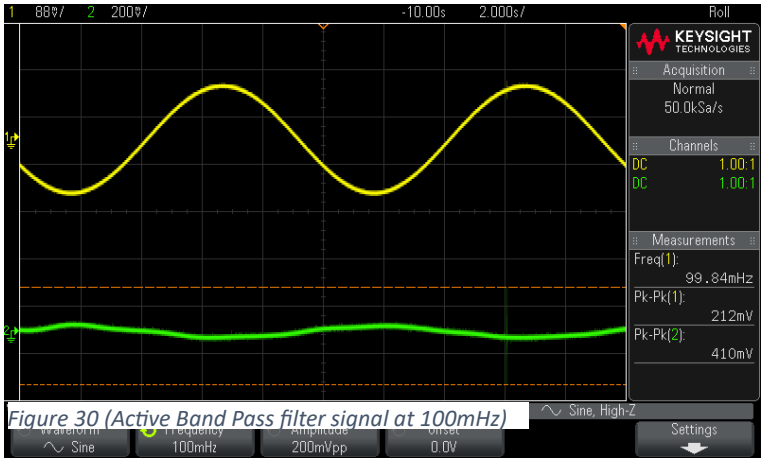


Figure 30 (Active Band Pass filter signal at 100mHz)

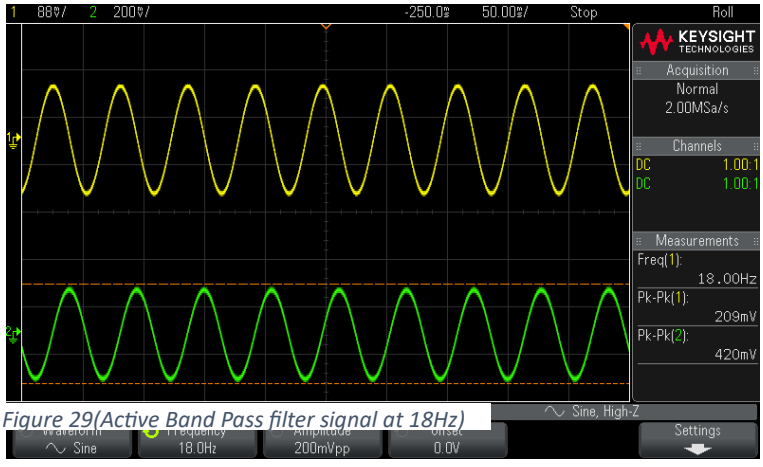


Figure 29(Active Band Pass filter signal at 18Hz)

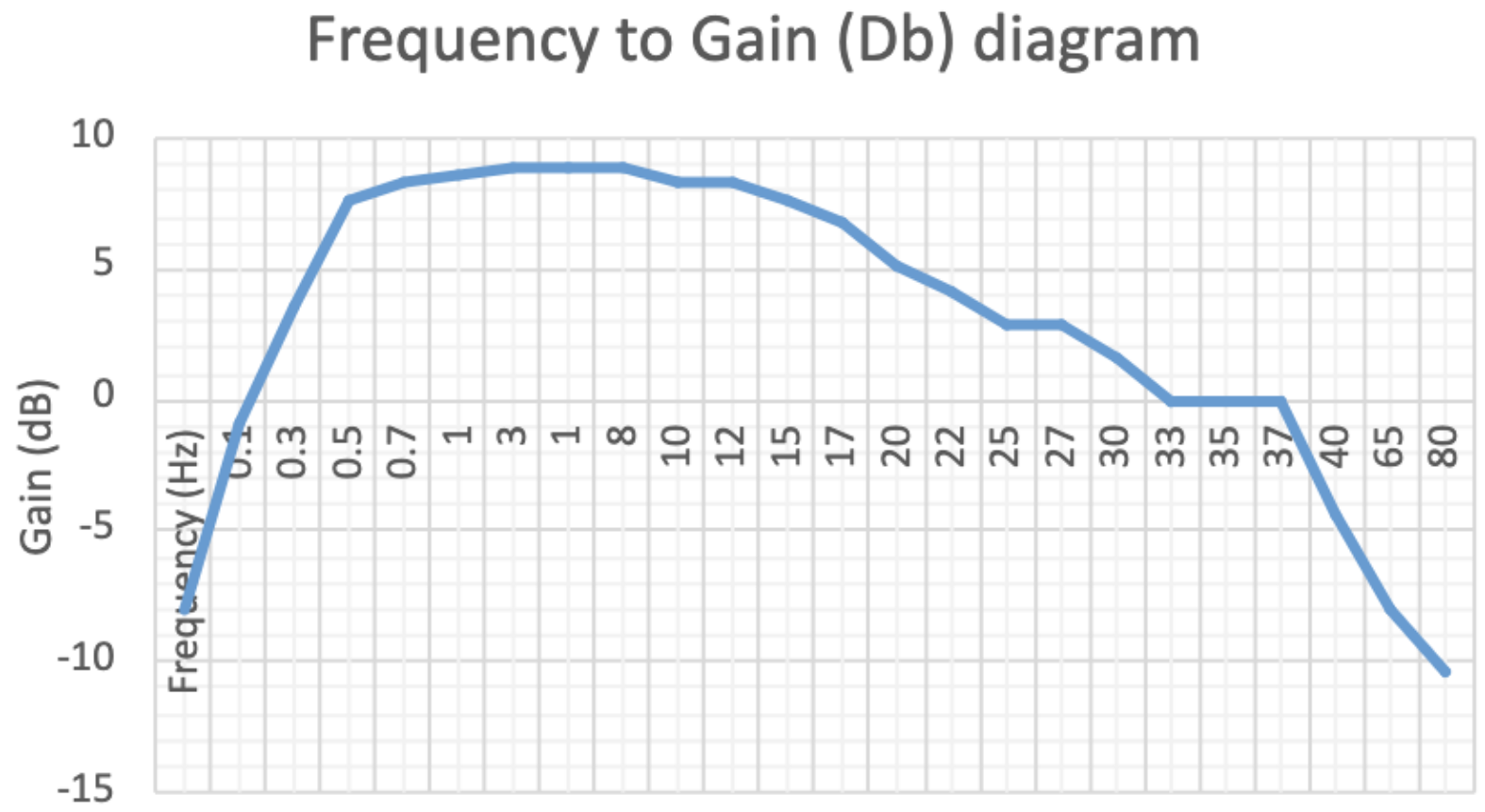


Figure 32(Frequency to Gain (dB) graph of Band pass filter)

Figure 31 (Table of Data for Band Pass Filter)

Frequency	Vin (mV)	Vout (volts)	Av	Gain (dBs) = 20log(Av)
0.1	200	80	0.4	-7.9588002
0.3	200	180	0.9	-0.9151498
0.5	200	300	1.5	3.52182518
0.7	200	480	2.4	7.60422483
1	200	520	2.6	8.29946696
3	200	540	2.7	8.62727528
1	200	560	2.8	8.94316063
8	200	560	2.8	8.94316063
10	200	560	2.8	8.94316063
12	200	520	2.6	8.29946696
15	200	520	2.6	8.29946696
17	200	480	2.4	7.60422483
20	200	440	2.2	6.84845362
22	200	360	1.8	5.1054501
25	200	320	1.6	4.08239965
27	200	280	1.4	2.92256071
30	200	280	1.4	2.92256071
33	200	240	1.2	1.58362492
35	200	200	1	0
37	200	200	1	0
40	200	200	1	0
65	200	120	0.6	-4.436975
80	200	80	0.4	-7.9588002
100	200	60	0.3	-10.457575

E. ECG Amplifier

An ECG amplifier built with a band-pass filter and an instrumentation amplifier on top creates a system that is precise in amplifying and isolating the electrical signals produced by the heart. The instrumentation amplifier is perfect for picking up weak heart signals since it has a high input impedance, minimal noise, and the capacity to reject undesirable signals. The signal is then further refined by eliminating noise outside of this range, such as electrical interference or muscle activity, by using a band-pass filter, which ensures that only frequencies within the heart rate range are passed through. Combined, they produce a magnified ECG signal that is clear enough for diagnostic use. We inputted a signal with an ECG simulator, replicating an ECG signal at millivolts. The following ECG signal was obtained (figure 36). Although it wasn't perfect amplification, the main characteristics of the PQRS waveform can be seen as intervals between each wave. This shows that the signal lies within our Pass band frequency and has been amplified accordingly by our instrumental amplifier. Proving the circuit we have built, the values selected for our capacitors and resistors were adequate, and how we've positioned our components to reduce noise as shown by minimal noise and clear output signal, as shown in Figure 35.

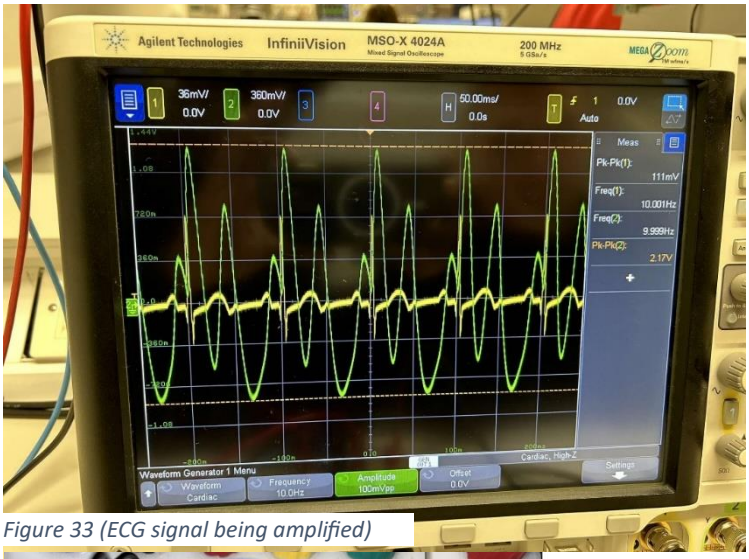


Figure 33 (ECG signal being amplified)

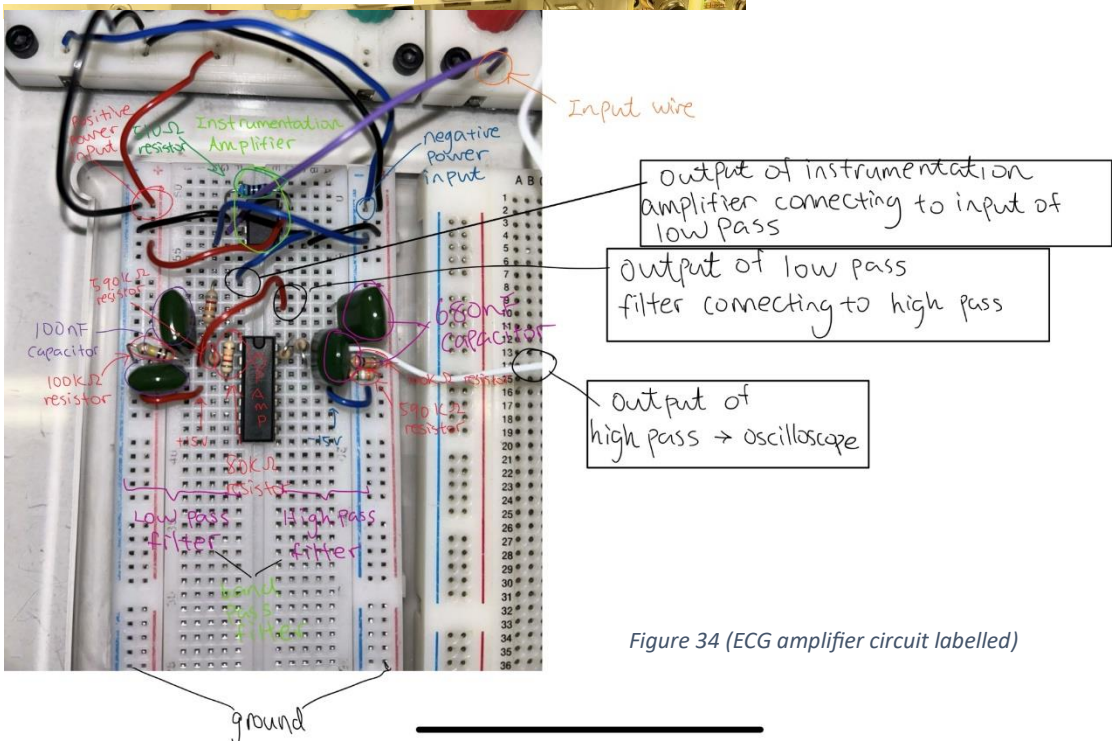


Figure 34 (ECG amplifier circuit labelled)

Part A discussion:
The operational amplifier performed as expected, inverting our input signal with minimal noise as our output signal wasn't noisy or distorted. However, when comparing our theoretical gain value to practical, there was a difference of -0.38, which decreased as frequency increased. This is due to Op-Amp non-idealities as functional circuits do not exhibit ideal behaviour, such as finite input impedance, non-zero output impedance, and bandwidth limitations, which can affect the gain.

Part B discussion:
Our differential amplifier performed well, reaching a value of 32.58 dB for its CMRR, proving it's functional. While analyzing at 50H,z, our differential gain is Gd = 17.7 V / 1.05 V = 16.8,6, 1.34 less than our theoretical gai,n quite a bit lower than we expected. As we increase our frequency from 50Hz to 1kHz, our signal has a higher differential gain than our previous frequency—a trend like our last inverting amplifier. Ideally, we would want the typical mode gain to be 0. However, there is often an associated common-mode gain, and in our trial, we obtained the value 0.396 dB. It is sufficient for noise rejection. Such a high common mode noise may be because in circuits where resistors are used to create a voltage divider, using different values can lead to incorrect biasing of components, potentially leading to increased susceptibility to noise and erratic operation.

Part C discussion:
Our instrumental amplifier performed well. With a CMRR value of 58.98, much higher than our differential amplifier. While there was a significant difference between our theoretical differential gain = 99, our practical gain was 92.52, a difference of 6.48. However, our common-mode gain was better, with only a difference of 0.104. Our instrumental amplifier performed well since we only needed to connect one resistor and wire it to the INA121 chip, which reduced a lot of noise as there were much fewer components required, reducing the factor of component tolerance meaning their actual values can differ from the nominal values by a certain percentage. This can affect the gain calculation, which depends on these component values and materials.

Part D discussion:
Our bandpass filter performed well, but it had minor differences in what we expected for its cut-off frequencies. This is due to using different resistors of different values in circuits; resistors are often used in pairs. If these resistors are not closely matched in value, this can lead to an imbalance in the circuit, which may cause it to amplify common-mode noise (noise present on both inputs).

Part E discussion:
Due to time constraints, we faced setbacks as we could not analyze the ECG signal without the bandpass filter. The amplification of the ECG signal was also different from what we had expected. One possible explanation for the reduced attenuation of frequency response could be that the order of our second-order filter was set too low. Because of the low input impedance of the circuit components, noise in the filter might not have been adequately filtered out. Alternatively, since Butterworth's filter is being used, the Butterworth wave is relatively flat, but other filters, like Chebyshev, have ripples before they slew off.

Finishing Remarks

The importance of instrumentation amplifiers for biopotential measurements (compared to simple non-inverting amplifiers).

High Input Impedance: Instrumentation amplifiers with an extremely high input impedance are essential for biopotential measurements like ECGs. They have elevated input impedance guarantees that the amplifier does not load the signal source, maintaining the integrity of the fragile biopotential signals. Note that instrumental amplifiers have higher input impedance than a simple non-inverting amplifier. (Cromwell, L., Weibell, F.J. and Pfeiffer, E.A., 1973)

Superior Common-Mode Rejection Ratio (CMRR): Instrumentation amplifiers can efficiently reject common-mode interference and noise, like power line noise, which affects both input leads. This skill is crucial to obtain distinct biopotential signals in a noisy setting. A simple noninverting amplifier cannot do so, leading to increased noise in the ECG signal.

Differential signal processing: Common-mode signals are common to both inputs and are rejected by instrumentation amplifiers, which measure the difference between two voltages. Maintaining their high input impedance for both inputs is essential for precise differential signal processing.

What modifications are required to use this circuit to measure EMG?

EMG signals (0 to 10,000Hz) are on a much higher frequency compared to ECG signals (0.05-150Hz) (P A Kyriacou. 2024) , which means a modification to the bandwidth of our current circuit to modify the values of capacitors and resistors in our filter circuit to shift the cut-off frequencies of both high and low-pass filters. In addition to that, EMG signals are very vulnerable to interference; additional filtering using such a notch filter may become necessary. Lastly, if we did use electrodes for our ECG circuits, we may need different electrodes as ECG electrodes are placed in specific locations. EMG measurements are placed over a muscle group being studied; therefore, different types of electrodes are necessary.

In addition, describe other means of reducing electrical interference in biomedical experiments. In the laboratory, what were the different causes of interference in your ECG recording? How could you have reduced them?

Electrical interference could come from the powerlines, monitors, computers, phones, and even the devices we've been using, which all cause electrical interference as they emit a specific frequency. We can reduce this electrical interference by being in a controlled environment, which reduces the number of electronic devices within the vicinity of our experiment, which could include the use of battery-powered devices or isolation transformers to power our circuits. More importantly, double-check for any loose wires and securely attach them as they can cause motion artifacts that affect the signal generated. Lastly, as mentioned above, we could use a notch filter that can filter out a certain known frequency, which can affect the result of our ECG amplifier.

Conclusion

In summary, this project has been a thorough investigation of the field of biomedical instrumentation through the design and development of an ECG amplifier employing an instrumentation amplifier and a band-pass filter. The experimentation demonstrated the critical function of every element in improving signal accuracy and clarity. Even though there was some variance between the theoretical and practical results, these differences provided important new information about the intricacies of real-world circuit functioning. The study demonstrated how crucial accuracy is when choosing components and designing circuits, especially for vital applications like medical diagnostics. All things considered, the experience has strengthened comprehension of the fundamentals of biological instrumentation and highlighted the difficulties and benefits of real-world application in this area.

Bibliography/References

Cromwell, L., Weibell, F.J. and Pfeiffer, E.A., 1973. Biomedical Instrumentation and Measurements. 2nd ed. Englewood Cliffs, NJ: Prentice-Hall.

Webb, A.G., 2018. Principles of Biomedical Instrumentation. Cambridge: Cambridge University Press.

Webster, J.G., 2010. Medical Instrumentation: Application and Design. 4th ed. Hoboken, NJ: John Wiley & Sons.

P A Kyriacou. 2024 Lecture slides Introduction to Bio signal slides